Arab academy for Science, Technology and Maritime Transport

Sheet (3)

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1. Unsigned

Convert the following decimal numbers to 8-bit unsigned integers:

1. $(41)_{10}$
2. $(80)_{10}$
3. $(125)_{10}$
4. $(250)_{10}$

2. Sign and Magnitude

1) The following are sign-and-magnitude binary numbers in an 8-bit allocation. Convert them to decimal:

1. $(10101101)_2$
2. $(01110111)_2$
3. $(11111100)_2$
4. $(01110100)_2$
5. $(11001110)_2$

2) Change the following decimal numbers to sign-and-magnitude binary numbers in an 8-bit allocation:

1. $(10)_{10}$
2. $(-7)_{10}$
3. $(-55)_{10}$
4. $(112)_{10}$
5. $(-25)_{10}$

3. Two's Complement

Change the following decimal numbers to 8-bit two's complement integers:

1. $(-5)_{10}$
2. $(80)_{10}$
3. $(-10)_{10}$
4. $(-125)_{10}$
5. $(120)_{10}$